

RANJAN KHADKA

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SUMMARY

Software Engineering student with hands-on experience developing and deploying full-stack web applications, backend integrations, Linux-based systems, AI-powered applications, and technical support solutions. Experienced leading student teams, collaborating on real-world sustainability projects, troubleshooting technical issues, and supporting campus operations in fast-paced environments. Founding member of a campus technology organization.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C++, SQL, HTML, CSS, C
Frameworks and Technologies: React, Node.js, Express.js, REST APIs, Vite, Nginx, Hadoop, Spark, HDFS
Databases: SQL, MySQL, MongoDB
Tools and Platforms: Git, GitHub, Linux, Unix, Red Hat Enterprise Linux, Railway, Vercel, Docker, AWS, JIRA, VS Code
Concepts: Full-Stack Development, API Integration, Database Design, OOP, SDLC, Debugging, Troubleshooting, Leadership

EDUCATION

University of Minnesota Crookston - Bachelor of Science in Software Engineering Expected Dec. 2027
Relevant Coursework: Data Structures and Algorithms, Software Engineering, Operating Systems, Database Systems, Computer Networks

EXPERIENCE

- Software Engineering Intern** | University of Minnesota Crookston Sustainability Department May 2025 - Present
- Developed and deployed UMC-Green, a sustainability-focused online marketplace where students can donate and receive items for free.
 - Worked on backend systems, API integrations, deployment workflows, database engineering, and server configuration.
 - Migrated UMC-Green from MongoDB to a SQL-based database on a Red Hat Enterprise Linux server.
 - Configured Linux-based hosting environments and managed web server deployment using Nginx, Railway, and Vercel.
 - Integrated authentication and university system compatibility features while collaborating in a two-person development team.
- IT Help Desk Technician** | University of Minnesota Crookston Oct. 2024 - Present
- Provide technical support and troubleshooting assistance for students and staff across campus systems.
 - Diagnose hardware, software, networking, account access, software setup, and device configuration issues.
 - Maintain professional, clear communication in customer support interactions.
- Student Leader** | University of Minnesota Crookston Facilities and Grounds May 2024 - Present
- Lead and coordinate student workers in daily facilities and grounds operations across campus.
 - Assist with task delegation, scheduling, workflow management, and safe completion of campus projects.
 - Recognized with the Facilities Student of the Year Award for reliability, leadership, teamwork, and campus contributions.
- Selected Community Advisor** | University of Minnesota Crookston Aug. 2026 - Apr. 2027
- Selected to support and mentor residential students in a diverse campus community.
 - Will assist with student problem-solving, conflict resolution, campus resources, and community-building activities.

PROJECTS

- 11Bunkers** | React, Node.js, Claude API, ElevenLabs API, Vercel, Railway eleven-bunkers.vercel.app
- Led a hackathon team that built an AI-powered interactive storytelling game and won 1st place at St. Cloud State University.
 - Integrated Claude API to dynamically generate story content and game scenarios.
 - Integrated ElevenLabs API to narrate generated stories and create an immersive game-master experience.
- UMC-Green Marketplace** | React, Node.js, SQL, Red Hat Enterprise Linux, Railway, Vercel, Nginx
- Built a full-stack sustainability platform enabling students to exchange reusable items for free.
 - Migrated the application from MongoDB to SQL and configured production deployment on a Red Hat Enterprise Linux server.
- Linux Kernel Module Project** | C, Linux, Kernel Programming, Proc File System
- Developed Linux kernel modules interacting with the /proc filesystem and implemented process and PID handling functionality.
- Scalable E-Commerce and Real-Time Analytics System** | Hadoop, Spark, Spark SQL, HDFS, Python
- Implemented data processing workflows to compute revenue analytics, top products, and active user metrics.
- Task Manager Web App** | React, Node.js, MongoDB, JavaScript
- Created a full-stack task management application with CRUD functionality, authentication, responsive UI, and data persistence.

LEADERSHIP, AWARDS, AND INVOLVEMENT

- 1st Place, St. Cloud State University Hackathon** 11Bunkers
- Founding Member** | Innovation and Technology Alliance, University of Minnesota Crookston 2024 - Present
- Facilities Student of the Year Award** | University of Minnesota Crookston 2025
- Inclusion and Belonging Excellence Award** | University of Minnesota Crookston 2024
- New International Student Orientation Volunteer** | University of Minnesota Crookston Fall and Spring